

STATISTICS 321: Practice Lab

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Purpose

This lab is designed to prepare you for upcoming quizzes and examinations by practicing the types of reasoning expected in this course. You may use AI tools, notes, textbooks, class handouts, R, and other course resources. The goal is not simply to obtain answers, but to understand what is being expected, why a method works, when a method is appropriate, when a method becomes inefficient, and what assumptions are required.

Expectations

Show reasoning clearly. Use probability notation appropriately. Support claims with mathematical arguments. Explain why methods work and when they apply. If you disagree with a solution, identify the first step where the reasoning fails.

Question 1: Choosing and Justifying a Strategy

For each scenario: (a) identify the approach you would recommend first; (b) write the first counting or probability expression you would begin with; (c) identify a second approach that would also produce a correct answer; (d) explain why your preferred approach is more efficient.

Available approaches: Explicit sample space enumeration, Outcome table, Complement rule, Combination counting, Union/intersection reasoning, Sequential multiplication, Symmetry argument.

- A.** Fair coin flipped 15 times. Determine $P(\text{at least one Head})$.
- B.** A player wins any game with probability 0.5. Six games are played. Let X be the number of wins. Determine $P(X = 4)$.
- C.** One card is drawn from a standard 52-card deck. Let $K = \text{King}$ and $H = \text{Heart}$. Determine $P(K \cup H)$.
- D.** Two cards are drawn without replacement. Determine $P(\text{two Aces})$.
- E.** Two fair six-sided dice are rolled. Determine $P(X_1 > X_2)$.

Follow-Up: Choose two scenarios. For each: (i) show how your alternate method would begin; (ii) explain why the alternate method still works mathematically; (iii) explain what feature makes the alternate method cumbersome; (iv) describe one small change that would make your preferred method less attractive.